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E-COMMERCE APPLICATION

An Industrial Internship Report

Submitted by

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in partial fulfilment for the award of the degree of

M. TECH (INTEGRATED)

In

SOFTWARE ENGINEERING



VIT[®]
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

**SCHOOL OF COMPUTER SCIENCE ENGINEERING
AND INFORMATION SYSTEMS**

MAY - 2024

DECLARATION BY THE CANDIDATE:

I hereby declare that the Industrial Internship report entitled “**E-COMMERCE SYSTEM**” submitted by me to Vellore Institute of Technology, Vellore in partial fulfilment of the requirement for the award of the degree of **M. Tech (Integrated) in Software Engineering** is a record of bonafide industrial training undertaken by me under the supervision of **SMARTBRIDGE**. I further declare that the work reported in this report has not been submitted and will not be submitted, either in part or in full, for the award of any other degree in this institute or university.

A handwritten signature in black ink, appearing to read 'R. Naveen R.', is centered within a light gray rectangular box.

Signature of the student

Name: NAVEEN R

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Abstract

This project presents *shopEZ*, a full-stack e-commerce platform designed to provide an accessible, efficient, and personalized online shopping experience for a diverse range of users. Built using the MERN (MongoDB, Express.js, React.js, Node.js) stack, shopEZ aims to overcome common limitations in existing e-commerce platforms, which often lack adaptability, customization, and seamless performance across varying user needs. By focusing on an intuitive, user-centered design, shopEZ addresses the preferences of multiple demographics, such as busy shoppers looking for rapid purchasing options, budget-conscious college students, and senior citizens who prefer a straightforward and reliable shopping experience. The platform incorporates advanced search functionality, personalized recommendations, and a streamlined checkout process, ensuring users can quickly find and purchase items of interest. For college students, shopEZ offers exclusive discounts, budget-friendly selections, and trending items, facilitating affordable shopping. For senior users, the platform provides a clear interface with accessible navigation and optimized delivery reliability, ensuring satisfaction and confidence in their purchases. Leveraging the MERN stack enables robust data management, dynamic front-end interactions, and efficient back-end performance, contributing to a scalable, responsive, and user-oriented solution. This project underscores the importance of targeted design and scalability in fostering user satisfaction and loyalty within the e-commerce space.

Keywords:

E-commerce Platform, MERN Stack, Online Shopping, User-Centric Design, Accessibility, Personalization, Scalable Web Solutions, Targeted User Experience

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1. Introduction

With the growth of online shopping, e-commerce platforms have become essential tools for accessing a broad range of products with unparalleled convenience. However, many existing platforms lack the adaptability and personalized features that various user demographics require. Issues such as complicated interfaces, overwhelming advertisements, and limited affordable options often hinder the online shopping experience, particularly affecting busy professionals, budget-conscious college students, and senior citizens. This project, *shopEZ*, aims to address these gaps by offering a user-friendly, highly accessible e-commerce platform built on the MERN stack (MongoDB, Express.js, React.js, and Node.js). This full-stack solution delivers efficient, intuitive interactions, ensuring a positive experience for diverse user groups. With targeted features for different demographics, *shopEZ* includes a streamlined shopping experience for professionals with limited time, affordable options and discounts for college students, and a simplified interface with dependable service for senior users. Leveraging the scalability and responsiveness of the MERN stack, *shopEZ* is designed to provide a reliable and enjoyable shopping experience, enhancing customer satisfaction and loyalty.

1.1 Problem Statement

In today's fast-paced environment, e-commerce platforms are essential for shoppers who value convenience, yet they often fall short in meeting the unique needs of diverse user demographics. For example, busy professionals often find e-commerce sites cluttered with advertisements and confusing layouts, making it difficult for them to shop quickly and efficiently. College students on a budget face another issue, as these platforms prioritize higher-margin items, often neglecting affordable options, limiting students' ability to access budget-friendly products. Additionally, senior citizens may struggle with complex interfaces and unreliable delivery timelines, which diminish their confidence in online shopping. As a result, existing e-commerce platforms frequently fail to provide a consistently positive and tailored shopping experience for different user demographics, leaving users frustrated and dissatisfied.

1.2 Motivation

The development of *shopEZ* is motivated by the evolving demand for convenient, accessible, and personalized online shopping. Shoppers today expect seamless interactions, quick checkout, and tailored recommendations, making it essential to build a platform that addresses these needs effectively. The rising trend in online shopping, driven by a broader demographic spectrum, creates an opportunity to capture a growing market by offering a platform designed for diverse customer preferences. Additionally, modern full-stack technologies provide the technical foundation to create an efficient, interactive, and scalable e-commerce platform that emphasizes a personalized experience. This project is driven by the vision of transforming the shopping experience into one that is more user-centered, accessible, and enjoyable for all.

1.3 Objective

The objective of this project is to develop a user-friendly and robust e-commerce platform, *shopEZ*, using the MERN (MongoDB, Express.js, React.js, Node.js) stack. This system is designed to enhance the online shopping experience for various user demographics, including busy professionals, college students, and senior citizens. By prioritizing usability, affordability, and reliability, *shopEZ* aims to provide a tailored shopping journey, leveraging modern web technologies to deliver a scalable and responsive solution.

1.3.1 Proposed System

The proposed *shopEZ* e-commerce system is a full-stack web application built using the MERN stack, featuring the following functionalities:

- **Simplified Shopping Process:** A clean, easy-to-navigate interface with minimal steps to help users browse and purchase products quickly, catering to time-constrained professionals.
- **Customized Recommendations:** User data stored in MongoDB enables personalized product recommendations, special offers, and discounts based on user preferences and browsing history.

- **Affordable and Accessible Options:** A dedicated section for budget-friendly and popular items, making it easier for college students and cost-conscious users to find affordable products.
- **Reliable Delivery Tracking:** Real-time tracking and notifications that provide updates on order status, ensuring a reliable service, especially important for senior users.
- **Scalability and Real-Time Updates:** Node.js and Express.js ensure a high-performing backend that can handle high volumes of traffic, while React.js provides a dynamic, real-time front end for an uninterrupted shopping experience.

1.3.2 Advantages of Proposed System

1. **Enhanced User Satisfaction:** Personalized recommendations, a streamlined interface, and efficient processing create a satisfying and smooth shopping experience.
2. **Support for Budget and Accessibility:** By featuring budget-friendly options, *shopEZ* supports college students and price-sensitive users, providing affordable choices that fit within tight budgets.
3. **Reliable Delivery Experience:** Real-time tracking and timely updates give senior citizens and other users confidence in dependable deliveries, ensuring a stress-free experience.
4. **Scalability and Performance:** Built on the MERN stack, the platform is designed to handle peak traffic and perform smoothly, ensuring responsive interactions and real-time updates.
5. **Improved Customer Engagement:** Targeted offers and recommendations encourage repeat shopping and loyalty by addressing user interests and preferences effectively.

2. Technologies Learnt

To build the *shopEZ* e-commerce platform, various technologies were studied and applied, each contributing to the platform's robust functionality and seamless performance:

- **MongoDB:** A NoSQL database at the core of the MERN stack, MongoDB enables flexible and scalable data storage. Skills acquired include data modeling, efficient CRUD operations, complex query generation, and indexing to ensure optimal database performance.
- **Express.js:** A minimalist Node.js framework that simplifies backend development. Key skills gained include setting up routes, managing middleware, and creating RESTful APIs for seamless communication between the client and server.
- **React.js:** A library for building dynamic UIs, essential for developing an interactive and responsive front end. Knowledge in component-based architecture, state and props management, and the use of hooks like `useState`, `useEffect`, and `useContext` helped in building a smooth user interface.
- **Node.js:** A server-side runtime environment for executing JavaScript, essential for backend logic. Skills include asynchronous programming, handling HTTP requests, and configuring modules, which enabled the creation of a scalable and efficient backend.
- **Axios:** An HTTP client used for seamless communication between front end and back end. Skills gained include handling API calls, managing responses, and handling errors for reliable client-server interactions.
- **Bootstrap:** A CSS framework used to create a modern, responsive design. Knowledge of the grid system, responsive components, and customization options allowed for a cohesive, user-friendly UI across various devices.
- **Git and GitHub:** Version control with Git and GitHub was critical for tracking project changes, collaborating, and maintaining backups. Key skills acquired include branching, committing, merging, and resolving conflicts for efficient project management.

By mastering these technologies, the project achieved a full-stack solution combining a responsive front end, efficient data management, and robust server performance, providing a scalable, feature-rich, and user-focused e-commerce experience.

- **Checkout:** Once users are satisfied with their cart, they proceed to checkout. This module collects payment details, billing and shipping addresses, and the selected delivery method. Users can apply discount codes or use loyalty points during checkout.
- **Order Confirmation & Tracking:** After an order is placed, users receive an order confirmation with order details, estimated delivery time, and tracking options. This module allows users to track their orders in real-time, receiving updates on the processing, shipping, and delivery stages.
- **Profile:** The profile module lets users manage their personal details, including updating contact information, password, and default addresses. It also allows users to view order history, manage payment methods, and save favorite items for easy reordering.
- **Admin Dashboard:** This module is accessible to administrators for managing the platform. Admins can monitor user accounts, manage product listings, oversee order processing, and handle payments. Additionally, they can review customer feedback, view analytics, and monitor service quality.
- **Seller Dashboard:** This module is for sellers to manage their listings, including adding, updating, or removing products. Sellers can also set stock availability, prices, images, and descriptions for each product and view sales performance metrics.

3.3 System Specification

3.3.1 Software Requirements

Frontend:

- **React.js:** For building dynamic, interactive user interfaces.
- **HTML5 & CSS3:** For structuring and styling the application.
- **JavaScript (ES6+):** For handling client-side logic.
- **React Router:** For navigating between different pages.
- **Axios:** For sending HTTP requests to the backend.

Backend:

- **Node.js:** For server-side application logic.
- **Express.js:** For creating RESTful APIs.
- **MongoDB:** A NoSQL database for storing user, product, and order data.
- **Mongoose:** For object modeling and interacting with MongoDB.
- **JWT:** For secure user authentication and session management.
- **Bcrypt:** For password encryption to enhance security.

Version Control & Deployment:

- Git: For version control.
- GitHub/GitLab: For code repository hosting and collaboration.

3.3.2 Hardware Requirements

Development Environment:

- Server: Intel i5 or equivalent processor, 8GB RAM, 100GB SSD storage.
- Client Devices: Android 5.0+, iOS 10.0+, Windows 10+, macOS 10.12+ for optimal compatibility.

Production Environment:

- Web Server: Quad-Core processor, 16GB RAM, 250GB SSD storage, 1Gbps bandwidth for handling high user traffic.
- Database Server: Hosted on MongoDB Atlas with flexible scaling options to accommodate data growth and demand.

3.4.Detailed Design

3.4.1. Use case Diagram

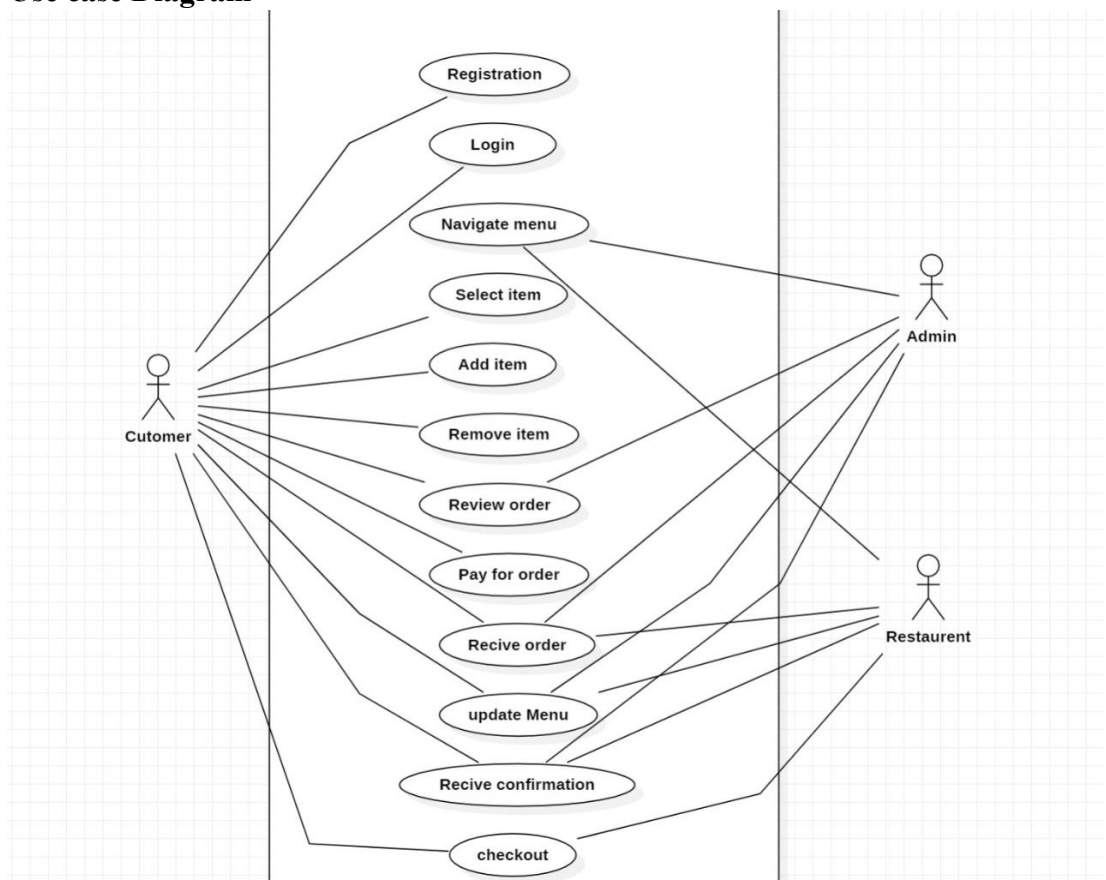


Fig 3.2 Use case Diagram

3.4.2. Sequence Diagram

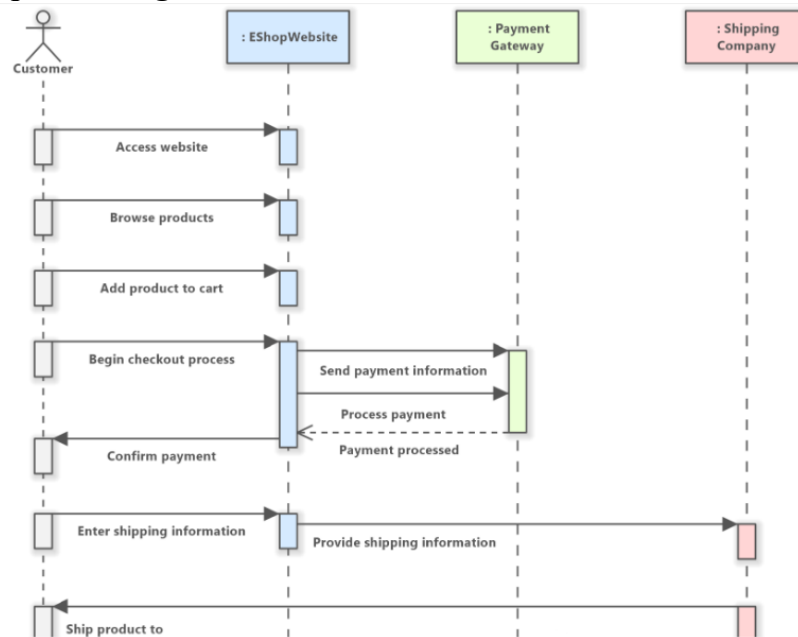


Fig 3.3 Sequence Diagram

3.4.3. Class Diagram



Fig 3.4 Class Diagram

3.4.4 Dataflow diagram

Level 0:

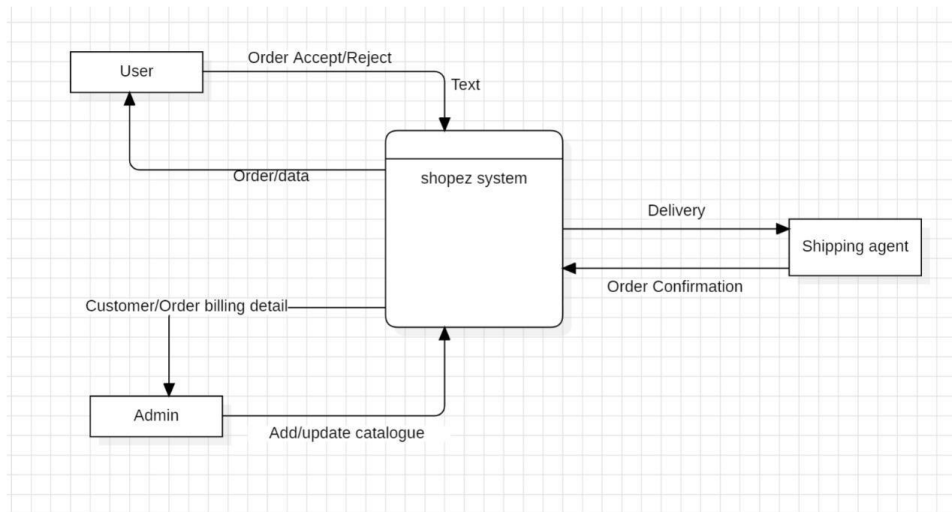


Fig 3.5 DFD Context diagram

Level 1:

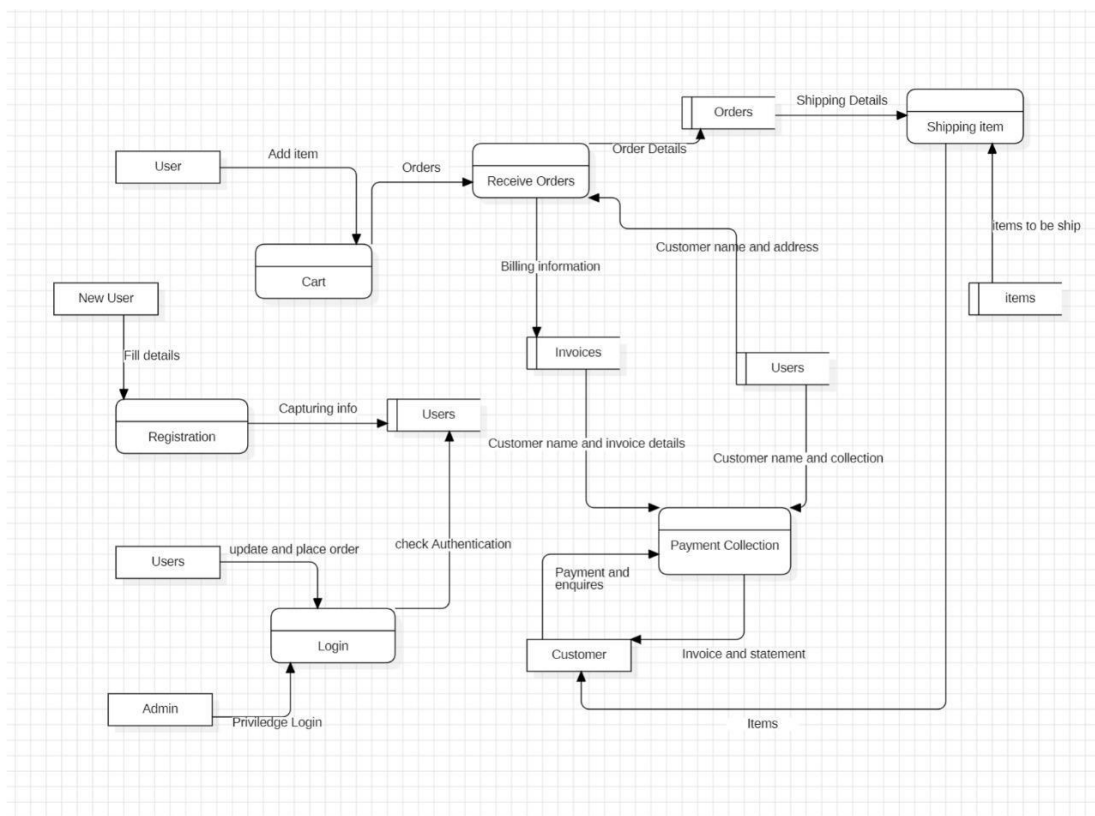


Fig 3.6 DFD Level 1 Diagram

3.4.5 Activity diagram

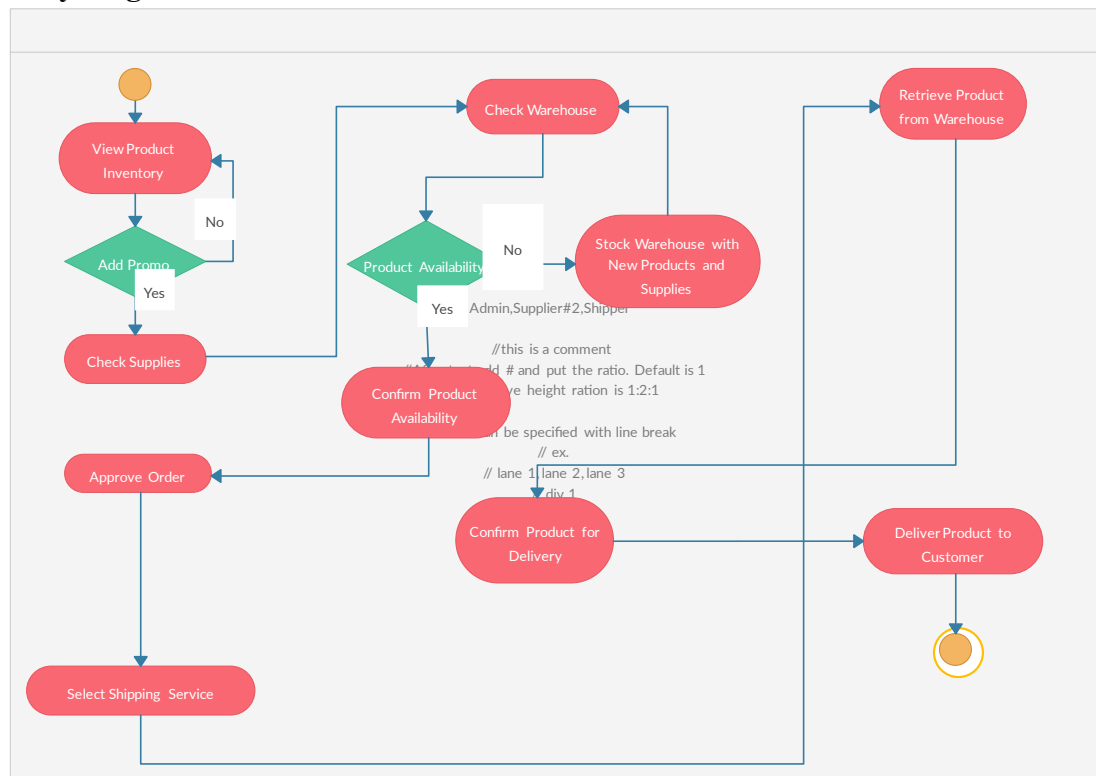


Fig 3.7 Activity Diagram

4. Implementation

4.1 Implementation details

The implementation of the E-Commerce System follows a structured approach involving both the frontend and backend development, with integration between the two to enable smooth operations. Below are the key implementation details:

SOURCE CODE:

<https://github.com/hrithikkb07/E-COMMERCE>

RESULTS:

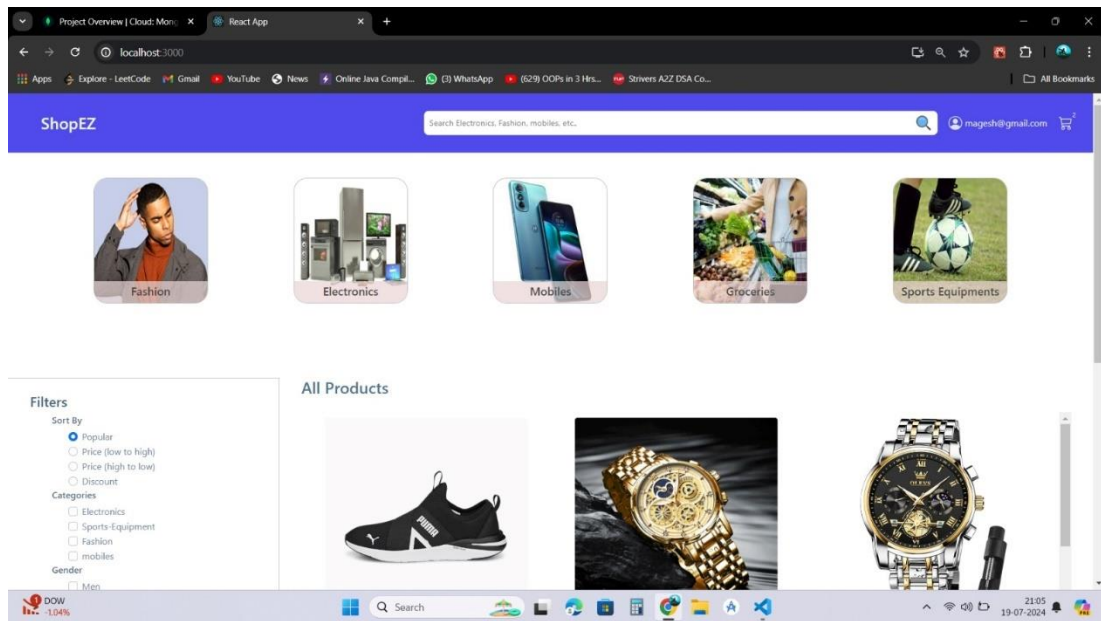


Fig 4.1 Home page

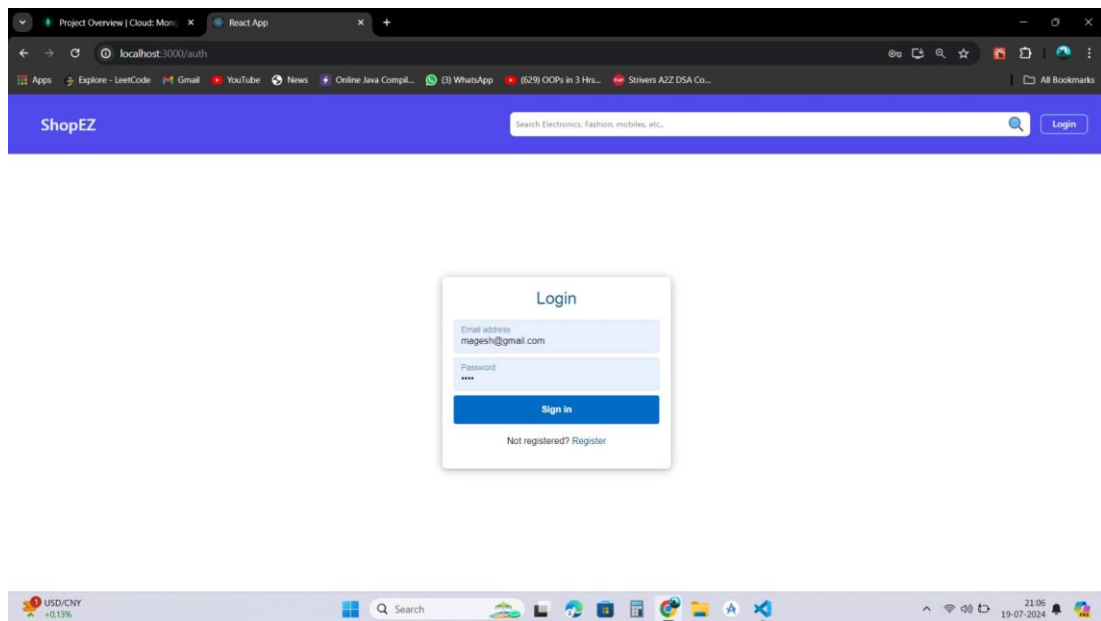


Fig 4.2 Login page

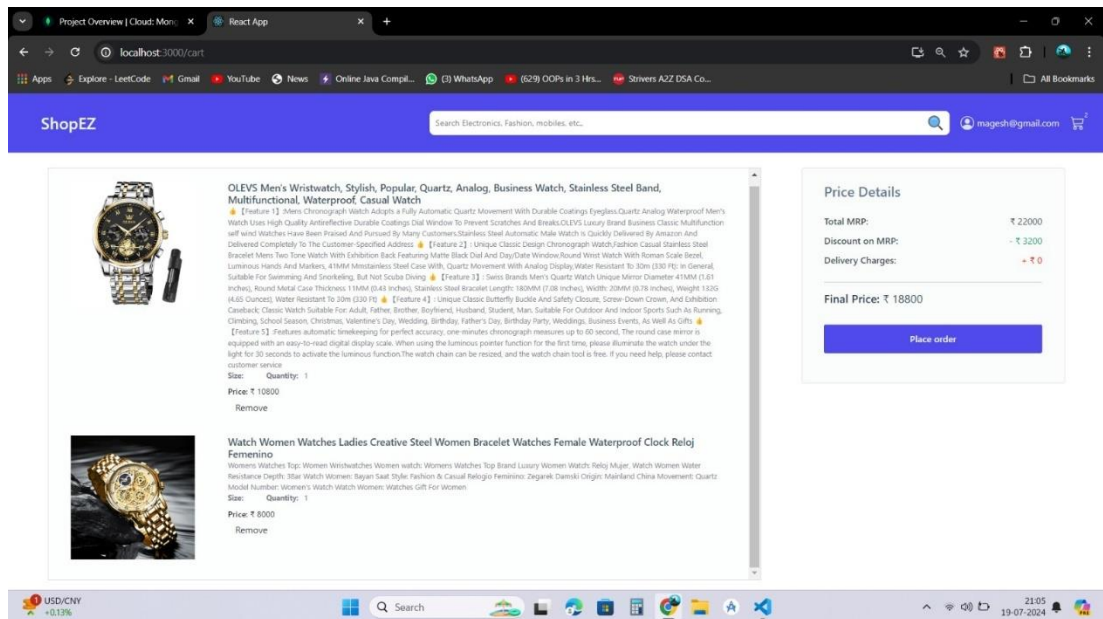


Fig 4.3 Cart page

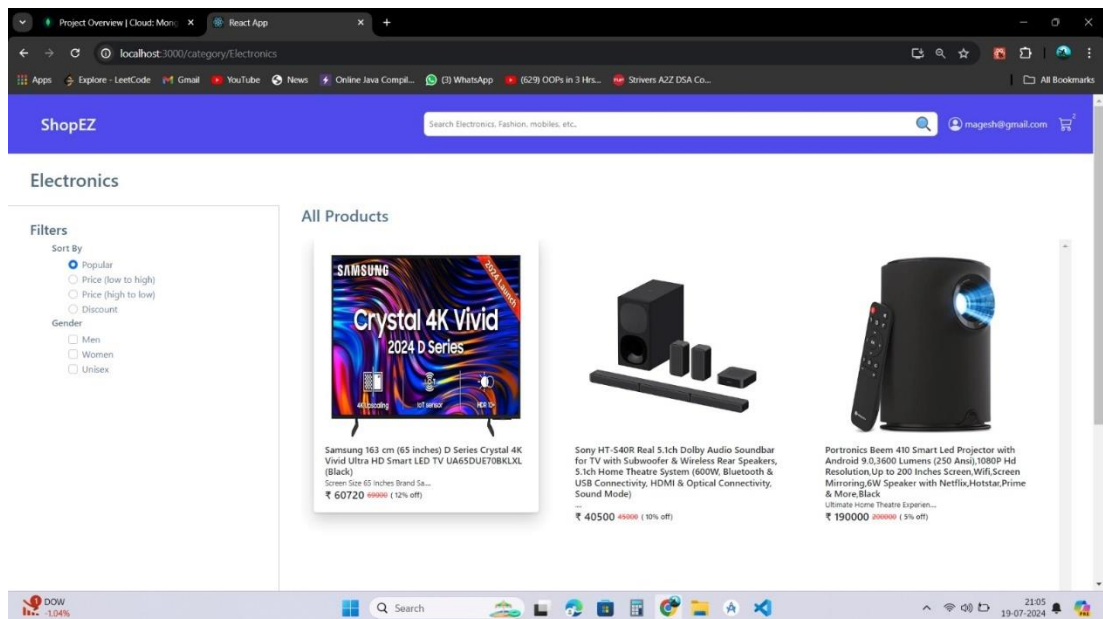


Fig 4.4 Products Page

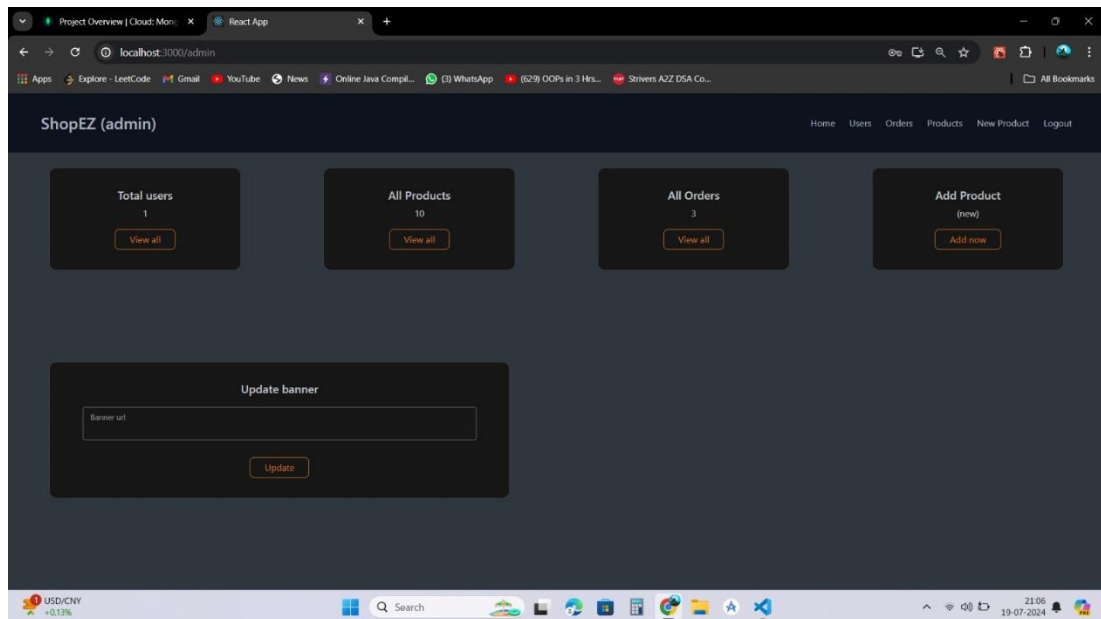


Fig 4.5 Admin Home page

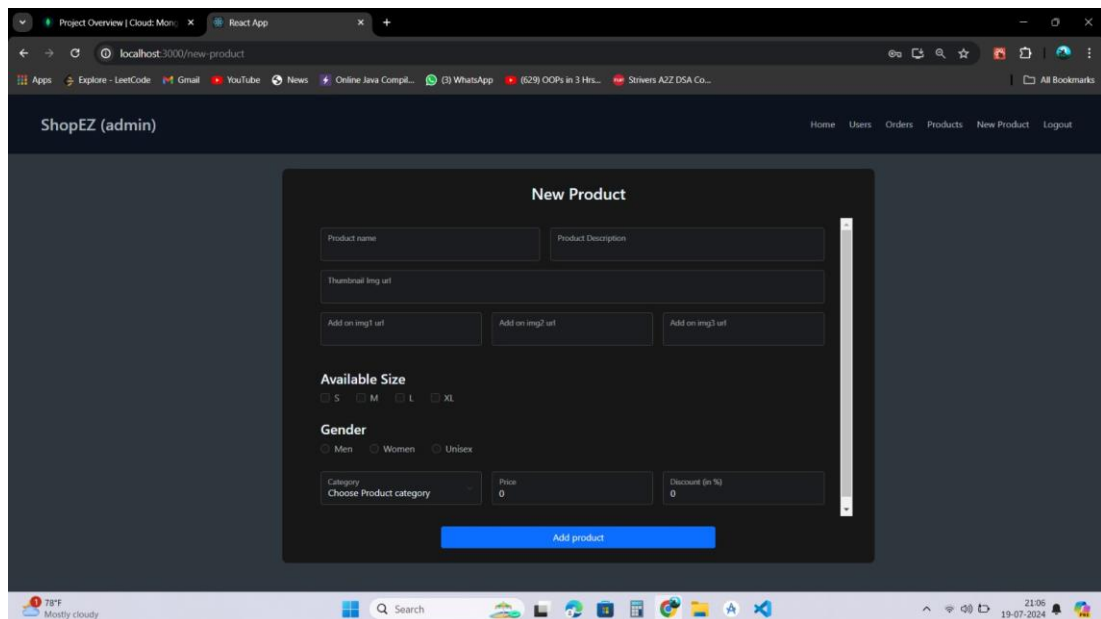


Fig 4.6 New Product add page

5. Test Results

5.1. Test cases

Functionalities that are being tested:

- User Login
- User Registration
- Browse Products
- Add to Cart
- Checkout
- Order History
- Logout

User Stories that are being tested:

- User entering valid Login credentials.
- User entering Invalid Login credentials.
- User registering a new account.
- Browsing products.
- Adding products to the cart.
- Checking out and completing the purchase.
- Viewing Order History.
- Logging out.

Testing Environment:

URL/Location: http://localhost:3000/

Credentials:

- Username: admin@example.com
- Password: admin123

Customer:

- Username: customer@example.com
- Password: customer123

Test Cases:

Table 5.1 testcases table

Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Pass/Fail
TC-001	User entering valid Login credentials.	1. Open the login page 2. Enter valid username and password 3. Click on login button	User should be logged in and redirected to the home page.	User logged in and redirected to home page	Pass
TC-002	User entering Invalid Login credentials.	1. Open the login page 2. Enter invalid username and password 3. Click on login button	Error message should be displayed: "Invalid credentials."	Error message displayed	Pass
TC-003	User registering a new account	1. Open the registration page 2. Enter valid details (username, email, password, etc.) 3. Click on register button	User should be registered successfully and redirected to login page.	User registered and redirected to login page	Pass
TC-004	Browsing products	1. Login with valid credentials 2. Navigate to products page 3. Browse products	User should be able to view products	User able to view products	Pass
TC-005	Adding products to the cart	1. Login with valid credentials 2. Navigate to products page.	Selected products should be added to the cart.	Products added to the cart	Pass

6. Results and Discussion

The development of the *shopEZ E-commerce Platform* using the MERN stack has resulted in a scalable, efficient, and user-centric shopping experience that meets the

needs of diverse demographics in the online retail space. By leveraging MongoDB, Express.js, React, and Node.js, the system facilitated a seamless front-to-back interaction, allowing for efficient inventory management, dynamic product displays, and secure transaction processing. User feedback praised the platform's intuitive and responsive interface, which provides an easy-to-navigate, customizable shopping experience for a wide range of users, including young adults, busy professionals, and senior citizens. Retail partners also benefited from simplified inventory management, sales analytics, and streamlined order processing, enhancing operational efficiency and data-driven decision-making.

Challenges such as handling real-time updates, managing peak loads, and ensuring data consistency were addressed through optimization techniques, scalable architecture, and periodic testing. Iterative improvements based on usability testing refined the interface, optimized back-end processes, and contributed to a smoother user experience. The project successfully met its objectives by delivering a reliable and feature-rich e-commerce solution while offering insights into the practical application of full-stack development. This solid foundation provides ample room for future innovations, such as AI-driven product recommendations and enhanced accessibility features, which promise to increase the platform's value in the competitive e-commerce market.

7. Conclusion and Future Work

The development and deployment of the *shopEZ E-commerce Platform* using the MERN stack have resulted in a highly effective and user-centered solution for today's e-commerce demands. The integration of MongoDB, Express.js, React, and Node.js has enabled the platform to offer users a streamlined experience for browsing products, placing orders, and tracking shipments, enhancing the overall shopping experience. Robust security features, including encrypted data handling and secure payment processing, have contributed to user trust, while retailers benefit from enhanced product management tools and sales insights to optimize their offerings. Despite initial challenges such as maintaining high performance under heavy traffic and ensuring data reliability, these issues were successfully mitigated through a scalable architecture and continuous user-focused refinements.

Looking forward, plans for *shopEZ* include implementing machine learning to provide personalized product recommendations, launching a mobile application to increase accessibility, expanding language options for broader user inclusivity, and further optimizing system scalability to handle increased user loads. This project not only addresses current user needs but also establishes a strong base for future growth and advancement within the e-commerce sector. The experience gained from this project has been invaluable, equipping the team with practical skills in full-stack development, scalable architecture, and responsive design—preparing us to tackle real-world technological challenges effectively.

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